

COMPUTER SCIENCE**PAPER 1 (THEORY)****Maximum Marks: 70****Time Allowed: Three Hours***Reading Time: Additional Fifteen Minutes***Instructions to Candidates**

1. You are allowed an additional fifteen minutes for only reading the question paper.
2. You must NOT start writing during the reading time.
3. This question paper has 12 printed pages.
4. It is divided into two parts: Part I and Part II.
5. It has eleven questions in all.
6. Part I is compulsory and has two questions.
7. While attempting Multiple Choice Questions in Part I, you are required to write only ONE option as the answer.
8. Part II is divided into three sections: A, B and C.
9. Each section in Part II has three questions. Any two questions have to be attempted from each section.
10. The intended marks for questions are given in brackets [].

Instruction to Supervising Examiner

1. Kindly read aloud the Instructions given above to all the candidates present in the examination hall.

PART I – 20 MARKS*Answer all questions.**While answering questions in this Part, briefly indicate your working and reasoning, wherever required.***Question 1**

- (i) According to the Principle of Duality, the Boolean equation $A \cdot (A' + B) + B' \cdot 1 = A + B'$ will be equivalent to: (*Application*) [1]
- (a) $(A + (A' \cdot B)) \cdot (B' + 0) = A \cdot B'$
 - (b) $(A' + (A \cdot B')) \cdot (B + 0) = A' \cdot B$
 - (c) $(A + (A' \cdot B)) \cdot (B' + 1) = A \cdot B'$
 - (d) $(A \cdot (A' + B)) + (B' \cdot 0) = A + B'$
- (ii) The complement of the Boolean expression $X \cdot Y' + X' \cdot Z$ is: (*Application*) [1]
- (a) $(X' + Y) \cdot (X + Z')$
 - (b) $(X + Y') \cdot (X' + Z)$
 - (c) $X' \cdot Y + X \cdot Z'$
 - (d) $(X' + Y) + (X + Z')$

- (iii) Given below are two statements marked Assertion and Reason. Read the two statements carefully and choose the correct option. [1]

Assertion: $p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge (q \Rightarrow p)$

Reason: $p \Leftrightarrow q$ is true only when both p and q are true.

(Analysis)

- (a) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.
 - (b) Both Assertion and Reason are true but Reason is not the correct explanation for Assertion.
 - (c) Assertion is true and Reason is false.
 - (d) Both Assertion and Reason are false.
- (iv) The Boolean equation $A \cdot (A + B) = A$ represents the: (Recall) [1]
- (a) Distributive law
 - (b) Absorption law
 - (c) Associative law
 - (d) Idempotent law
- (v) The canonical SOP form of $F(A, B, C) = \Sigma(2, 3, 7)$ is: (Application) [1]
- (a) $A' \cdot B \cdot C' + A' \cdot B \cdot C + A \cdot B \cdot C$
 - (b) $A \cdot B' \cdot C + A \cdot B' \cdot C' + A' \cdot B' \cdot C'$
 - (c) $(A + B' + C) \cdot (A + B' + C') \cdot (A' + B' + C')$
 - (d) $A' \cdot B \cdot C' + A' \cdot B \cdot C + A \cdot B' \cdot C$
- (vi) The complexity of the following code segment is: [1]

```
int sum = 0;
for (int i = n; i > 0; i = i / 2)
    sum = sum + i;
```

(Analysis)

- (a) $O(n)$
- (b) $O(\log n)$
- (c) $O(n^2)$
- (d) $O(1)$

(vii) Consider the following statement:

[1]

```
String s = "COMPUTER";
```

Which of the following statements are valid?

I. `s.substring(3)` returns "PUTER".

II. `s.indexOf('T')` returns 5.

III. `s.charAt(8)` returns 'R'.

(Application)

(a) I and II

(b) II and III

(c) I and III

(d) I, II and III

(viii) Given below are two statements marked Assertion and Reason. Read the two statements carefully and choose the correct option.

[1]

Assertion: In a switch statement, if the break statement is omitted after a case, the statements of the following case(s) are also executed.

Reason: The break statement transfers control out of the switch block.

(Analysis)

(a) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.

(b) Both Assertion and Reason are true but Reason is not the correct explanation for Assertion.

(c) Assertion is true and Reason is false.

(d) Both Assertion and Reason are false.

(ix) Write the converse of the statement: "If the battery of the phone is low, then the phone switches to power-saving mode." *(Application)*

[1]

(x) Differentiate between the == operator and the equals() method when two String objects are compared. *(Understanding)*

[1]

Question 2

(i) Convert the following infix notation to prefix form:

[2]

$(P - Q) * (R + S / T) - U$ *(Create)*

(ii) A two-dimensional array `arr[12][20]` (12 rows and 20 columns) is stored in the memory column major wise. If the address of `arr[1][2]` is 3000 and the address of `arr[7][9]` is 3360, find the number of bytes *W* occupied by each element. *(Application)*

[2]

- (iii) The following function `search()` is a part of some class. Assume `a[]` is an array of integers, `lo` is its lower bound and `hi` its upper bound. Answer the questions given below along with dry run / working.

```
int search(int a[], int key, int lo, int hi)
{
    if (lo > hi)
        return -1;
    int mid = (lo + hi) / 2;
    if (a[mid] == key)
        return mid;
    else if (a[mid] < key)
        return search(a, key, mid + 1, hi);
    else
        return search(a, key, lo, mid - 1);
}
```

- (a) What will the function `search()` return if `a[] = {3, 8, 15, 21, 30, 42, 57}`, `key = 42`, `lo = 0` and `hi = 6`? (*Analysis*) [2]
- (b) What is the function `search()` performing apart from recursion? (*Analysis*) [1]
- (iv) The following recursive function `isPal()` is a part of some class. It returns true if the part of the string `s` from index `f` to index `l` reads the same from both ends, otherwise false. It is invoked as `isPal(s, 0, s.length() - 1)`. [Example: MADAM is a palindrome.]

There are some places in the code marked by ?1?, ?2?, ?3? which may be replaced by a statement/expression so that the function works properly.

```
boolean isPal(String s, int f, int l)
{
    if (f >= l)
        return ?1?;
    if (s.charAt(f) != ?2?)
        return false;
    return ?3?;
}
```

- (a) What is the expression or statement at ?1? (*Analysis*) [1]
- (b) What is the expression or statement at ?2? (*Analysis*) [1]
- (c) What is the expression or statement at ?3? (*Analysis*) [1]

PART II – 50 MARKS

Answer six questions in this part, choosing two questions from Section A, two from Section B and two from Section C.

SECTION A

Answer any two questions.

Question 3

- (i) An online examination portal allows a candidate to log in (X) if any one of the following conditions is satisfied: [5]

- The candidate is registered, has paid the fee and is not late.

OR

- The candidate is registered, has downloaded the admit card and is not late.

OR

- The candidate is registered, has paid the fee and has downloaded the admit card, even if late.

The inputs are:

INPUTS

- | | |
|---|--|
| R | The candidate is registered. |
| F | The candidate has paid the examination fee. |
| A | The candidate has downloaded the admit card. |
| L | The candidate is late by more than 10 minutes. |

(In all the above cases, 1 indicates yes and 0 indicates no.)

Output: X – denotes that the candidate is allowed to log in [1 indicates yes and 0 indicates no in all cases].

Draw the truth table for the inputs and outputs given above and write the POS expression for $X(R, F, A, L)$.

(Application)

- (ii) Reduce the above expression $X(R, F, A, L)$ by using a 4-variable Karnaugh map, showing the various groups (i.e. octal, quads and pairs). Draw the logic gate diagram for the reduced expression using NOR gates only. Assume that the variables and their complements are available as inputs. (Create) [5]

Question 4

- (i)
- (a) Reduce the Boolean function $F(A, B, C, D) = \Sigma(1, 3, 4, 5, 6, 7, 9, 11, 12, 13)$ by using a 4-variable Karnaugh map, showing the various groups (i.e. octal, quads and pairs). (Application) [4]
- (b) Draw the logic gate diagram for the reduced expression using NAND gates only. Assume that the variables and their complements are available as inputs. (Create) [1]

(ii) Answer the following questions related to the image given below:



- (a) What is the output of the above gate if $A = 1$ and $B = 0$? (*Analysis*) [1]
- (b) What are the values of the inputs if the output is 1? (*Application*) [1]
- (iii) Verify if the following proposition is valid using the truth table: [3]
 $X \Rightarrow (Y \vee Z) = (X \Rightarrow Y) \wedge (X \Rightarrow Z)$ (*Application*)

Question 5

- (i)
- (a) What is an encoder? (*Recall*) [1]
- (b) Draw the logic circuit of a decimal to binary (BCD) encoder for the decimal digits 0 to 9. (*Application*) [3]
- (c) State one difference between an encoder and a decoder. (*Understanding*) [1]
- (ii) Draw the logic circuit of a full adder using two half adders and an OR gate. Write the Boolean expressions for its SUM and CARRY outputs. (*Create*) [3]
- (iii) Draw the logic circuit of a 2-input XOR gate using NOR gates only. Show the expression at the output of each gate. (*Create*) [2]

SECTION B

Answer any two questions.

Each program should be written in such a way that it clearly depicts the logic of the problem.

This can be achieved by using mnemonic names and comments in the program.

(Flowcharts and Algorithms are not required.)

The programs must be written in Java.

Question 6

A class LCM has been defined to find the Lowest Common Multiple of two integers.

[10]

Example: INPUT : n1 = 12, n2 = 18

OUTPUT : LCM = 36

Some of the data members and member functions are given below:

Class name : **LCM**

Data members/instance variables:

n1 : to store an integer number

n2 : to store an integer number

large : integer to store the larger of n1 and n2

sm : integer to store the smaller of n1 and n2

l : to store the LCM of the two numbers

Methods/Member functions:

LCM() : default constructor to initialise the data members with legal initial values

void accept() : to accept n1 and n2, and to find large and sm

int getLCM() : returns the LCM of n1 and n2 using the **recursive technique**

void display() : to print the numbers n1, n2 and their LCM

Specify the class LCM giving details of the constructor(), void accept(), int getLCM() and void display(). Define a main() function to create an object and call the member functions accordingly to enable the task.

(Create)

Question 7

A class OddEven contains an array of integers. Using the details given below, form a new array from two such arrays which contains first all the odd numbers of the first array, then all the odd numbers of the second array, followed by all the even numbers of the first array and then all the even numbers of the second array, each in their original order.

[10]

Example: First array : { 12, 7, 9, 4, 15 }

Second array : { 3, 10, 8, 21 }

Resultant : { 7, 9, 15, 3, 21, 12, 4, 10, 8 }

The details of the members of the class are given below:

Class name : **OddEven**

Data members/instance variables:

arr[] : integer array

size : to store the number of elements

Methods/Member functions:

OddEven(int n) : parameterised constructor to initialise size = n and create the array

void input() : to accept the elements of the array

OddEven arrange(OddEven P, OddEven Q) : returns an object whose array holds the odd numbers of P and Q followed by their even numbers, as described above

void display() : to display the array elements

Specify the class OddEven giving details of the constructor(int), void input(), OddEven arrange(OddEven, OddEven) and void display(). Define a main() function to create objects and call the functions accordingly to enable the task.

(Create)

Question 8

A class PalWord has been defined to display those words of a sentence which are palindromes and whose length is a prime number. The words of the sentence are separated by a single blank space and the sentence is terminated by '.', '?' or '!'. [10]

Example 1: INPUT : MOM AND DAD WENT TO SEE MADAM AT NOON.

OUTPUT : MOM DAD MADAM

NUMBER OF SUCH WORDS : 3

Example 2: INPUT : NOON IS A GOOD TIME.

OUTPUT : NO SUCH WORD

(NOON is a palindrome of length 4, and A has length 1; neither length is prime.)

Some of the members of the class are given below:

Class name : **PalWord**

Data members/instance variables:

sent : to store the sentence in UPPER CASE

Methods/Member functions:

PalWord(String s) : parameterised constructor to assign s, converted to upper case, to sent

boolean isPalin(String w) : returns true if the word w is a palindrome, otherwise false

boolean isPrime(int n) : returns true if n is a prime number, otherwise false

void display() : displays the palindromic words of prime length and their number, by invoking isPalin() and isPrime(); displays "NO SUCH WORD" if there is none

Specify the class PalWord giving details of the constructor(String), boolean isPalin(String), boolean isPrime(int) and void display(). Define a main() function to create an object and call the functions accordingly to enable the task.

(Create)

SECTION C

Answer any two questions.

Each program should be written in such a way that it clearly depicts the logic of the problem stepwise.

This can be achieved by using comments in the program and mnemonic names or pseudo codes for algorithms. The programs must be written in Java and the algorithms must be written in general / standard form, wherever required / specified.

(Flowcharts are not required.)

Question 9

- (i) A railway reservation counter keeps a waiting line of at most 20 passengers. A passenger joins the line at the rear and the passenger at the front is served first. [4]

Define a class Waitlist with the following details:

Class name : **Waitlist**

Data members/instance variables:

name[] : array to hold the names of the passengers
cap : stores the maximum capacity of the array
front : to point to the index of the front
rear : to point to the index of the rear

Methods/Member functions:

Waitlist(int max) : constructor to initialise cap = max, front = rear = 0 and to create the array
void join(String n) : adds the name n at the rear if possible, otherwise displays the message "COUNTER FULL"
String serve() : removes and returns the name at the front; if the line is empty, returns "EMPTY"
void show() : displays the names in the line

Specify the class Waitlist giving details of the functions void join(String) and String serve().

Assume that the other functions have been defined.

The main() function and algorithm need NOT be written.

(Create)

- (ii) Name the entity described above and state its principle. *(Understanding)* [1]

Question 10

A bank pays compound interest, compounded annually, on a fixed deposit. The maturity amount is given by

[5]

$$\text{Amount} = P \times (1 + R/100)^N$$

where P is the principal, R the rate of interest per cent per annum and N the number of years; the interest is Amount – P. Both are rounded to two decimal places.

A super class Deposit has been defined to store the details of a deposit. Define a sub class Maturity to calculate the interest and the maturity amount. The details of the members of both the classes are given below:

Class name : **Deposit**

Data members/instance variables:

name : to store the name of the depositor
principal : to store the amount deposited (in decimals)
rate : to store the rate of interest per cent per annum (in decimals)

Methods/Member functions:

Deposit(...) : parameterised constructor to assign values to the data members
void show() : to display the deposit details

Class name : **Maturity**

Data members/instance variables:

years : to store the number of years
amount : to store the maturity amount (in decimals)
interest : to store the compound interest (in decimals)

Methods/Member functions:

Maturity(...) : parameterised constructor to assign values to the data members of both the classes
void compute() : to calculate the maturity amount and the compound interest as given above
void show() : to display the deposit details along with the number of years, the interest and the maturity amount

Assume that the super class Deposit has been defined. Using the concept of inheritance, specify the class Maturity giving the details of the constructor(...), void compute() and void show().

The super class, main function and algorithm need NOT be written.

(Create)

Question 11

- (i) A linked list is formed from the objects of the class Book. The class structure of the Book is given below: [2]

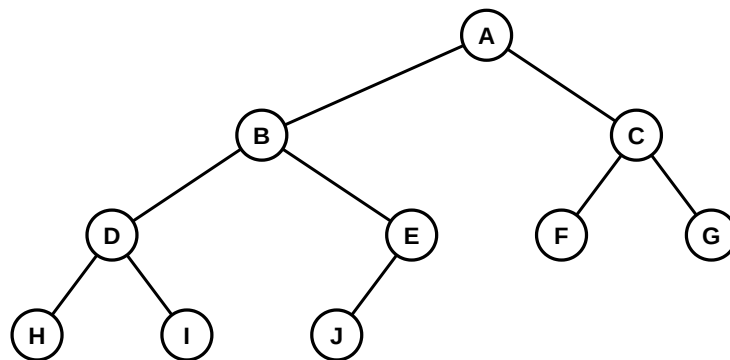
```
class Book
{
    String title;
    int price;
    Book next;
}
```

Write an Algorithm OR a Method to count and return the number of books in an existing linked list whose price is more than p. The method declaration is as follows:

`int countCostly(Book start, int p)`

(Analysis)

- (ii) Answer the following questions from the diagram of a Binary Tree given below:



- (a) Is the given tree a complete binary tree? Is it a full binary tree? Justify your answers. [1]
(Analysis)
- (b) Write the preorder traversal of the tree. *(Understanding)* [1]
- (c) State the degree of node E and name the external nodes of the tree. *(Understanding)* [1]

— — — —